

Open Science in Practice: Collaborative Research with Git



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Git basics

Andreas Gerhardsson

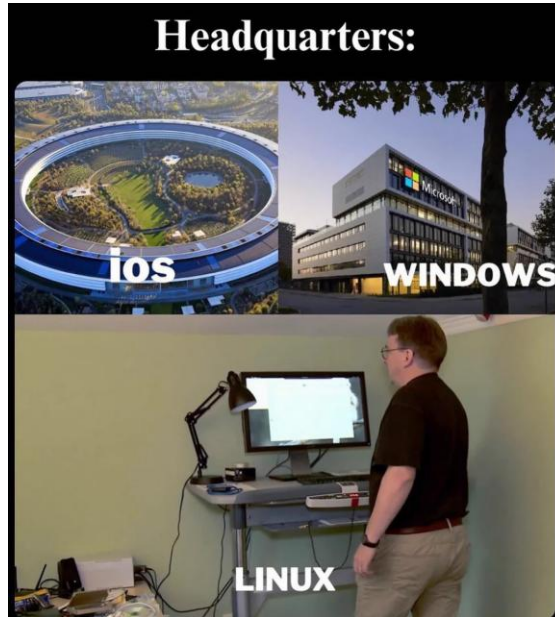
Why Git?



```
R_full_mlm.R
R_full_mlm2.R
R_full_mlm2old_2018_2.R
R_full_mlm2old_2018_corr.R
R_full_mlm2old_2018.R
R_full_mlm2old.R
R_full_mlm201710.R
R_full_nback_analysis.R
```

Git short (version) history

- A Distributed Version Control System (DVCS)
- Linus Torvalds (Linux) 2005
- Git $\neq$ Github



How does git work



Project



read_data.R



run_models.R



make_figures.R



export.R

```
andreas ~ %
```

How does git work



Project



read_data.R



run_models.R



make_figures.R



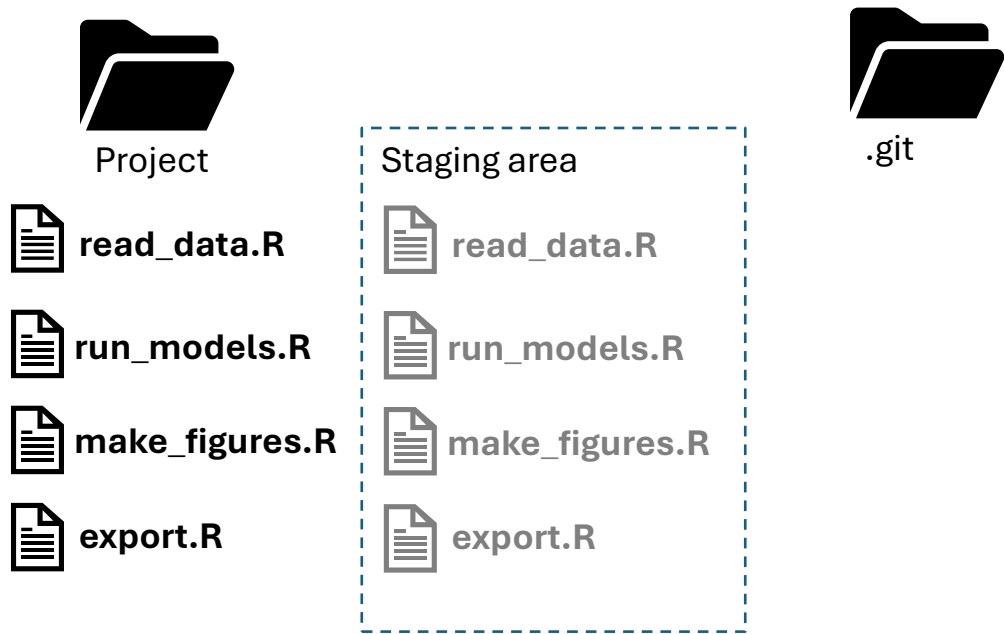
export.R



.git

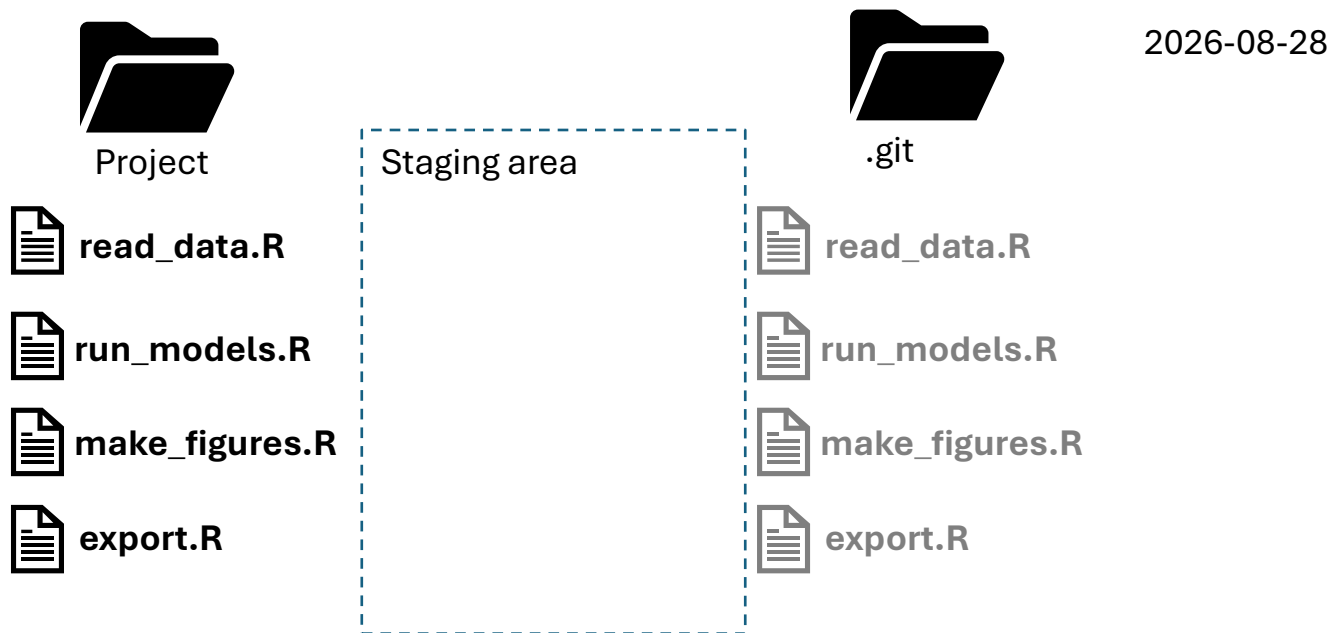
```
andreas ~ % git init
```

How does git work



```
andreas ~ % git add .
```

How does git work

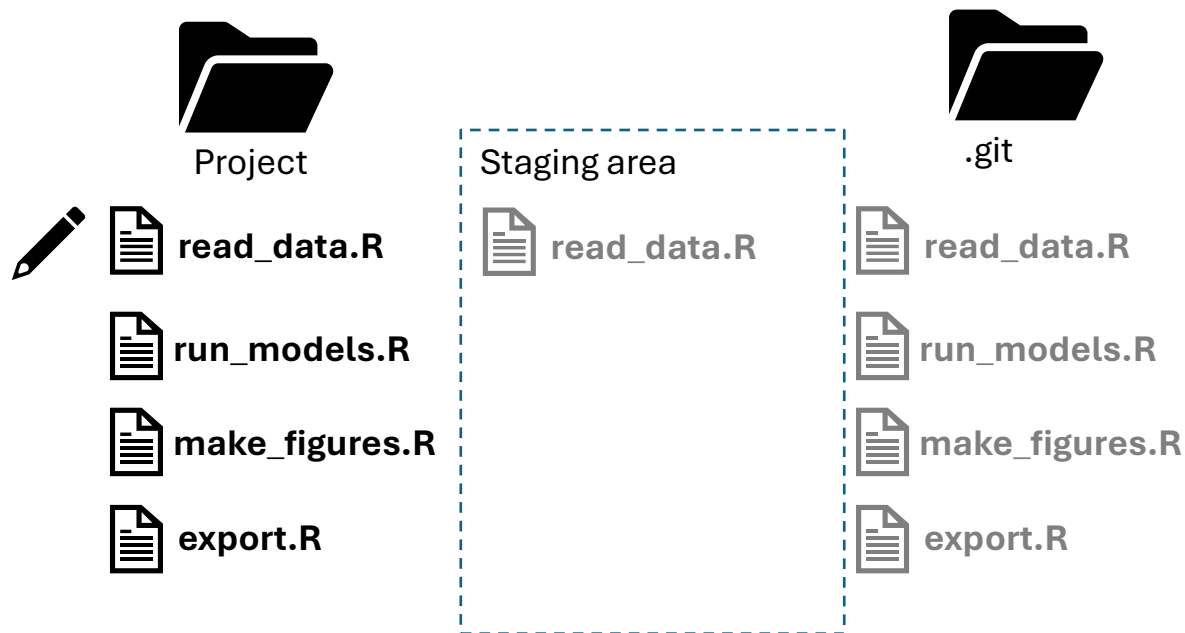


2026-08-28

Commit 1

```
andreas ~ % git commit -m "first commit"
```

How does git work



2026-08-28

Commit 1

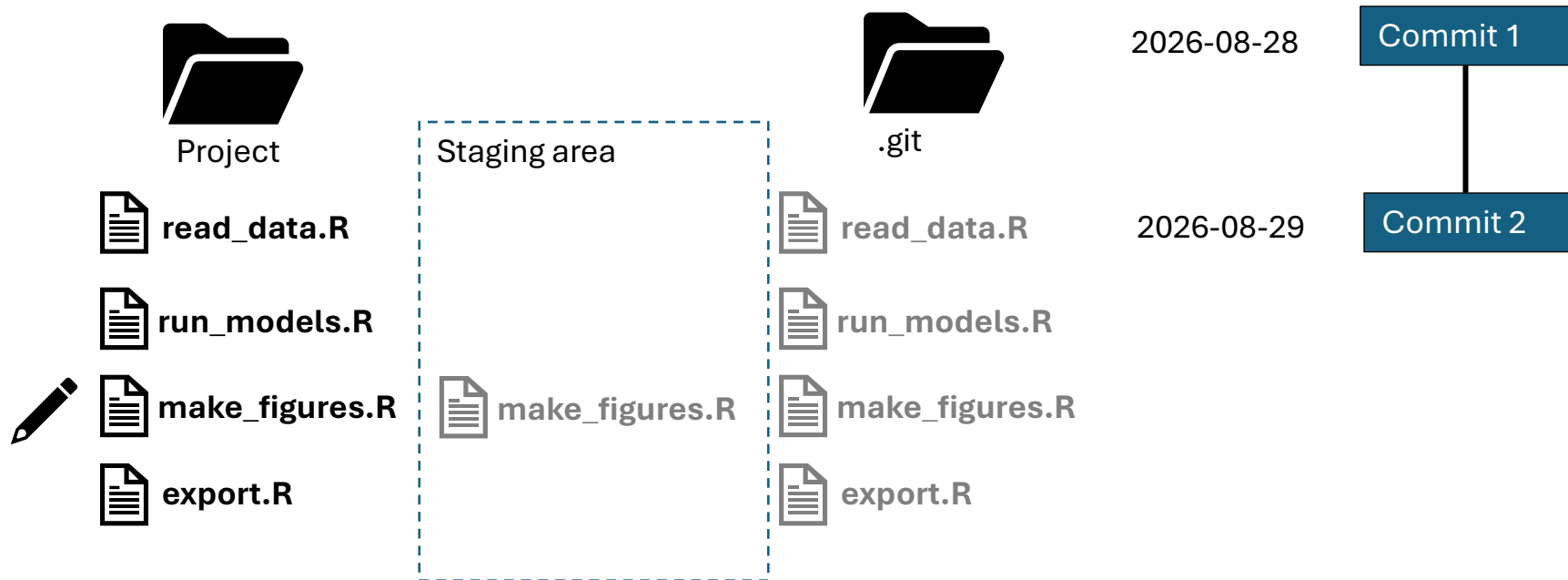
```
andreas ~ % git add read_data.R
```

How does git work



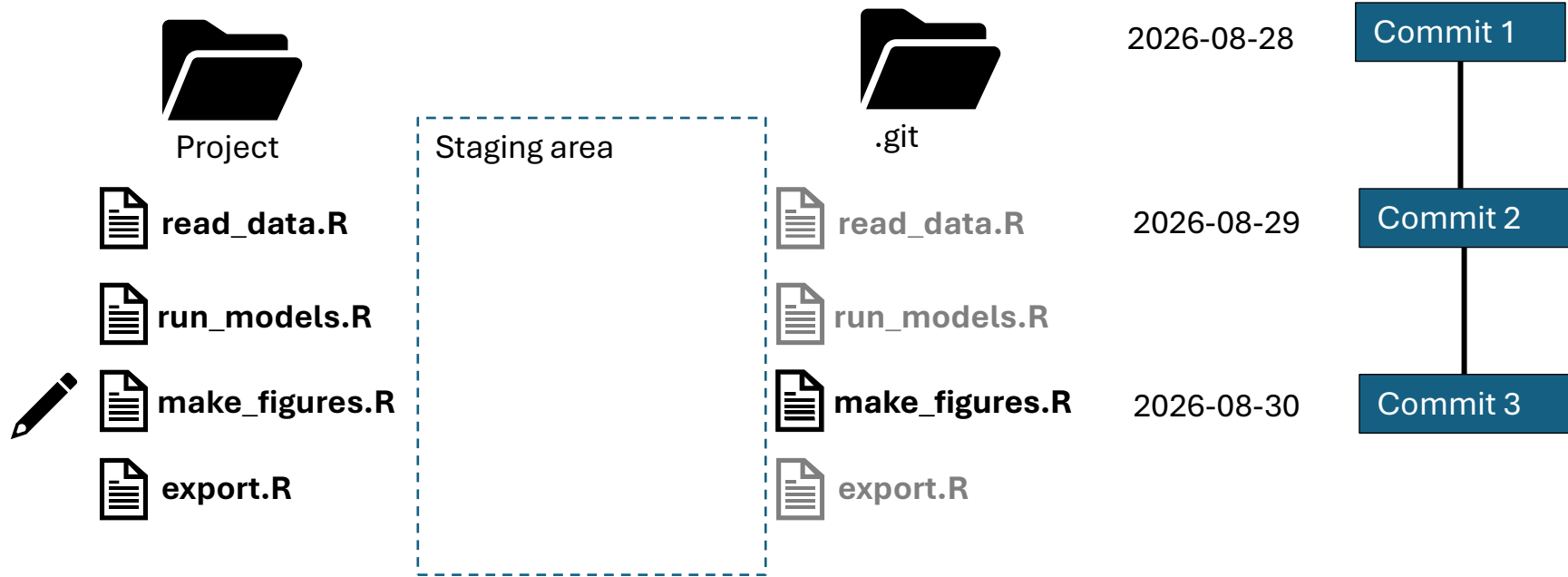
```
andreas ~ % git commit -m "fix path"
```

How does git work



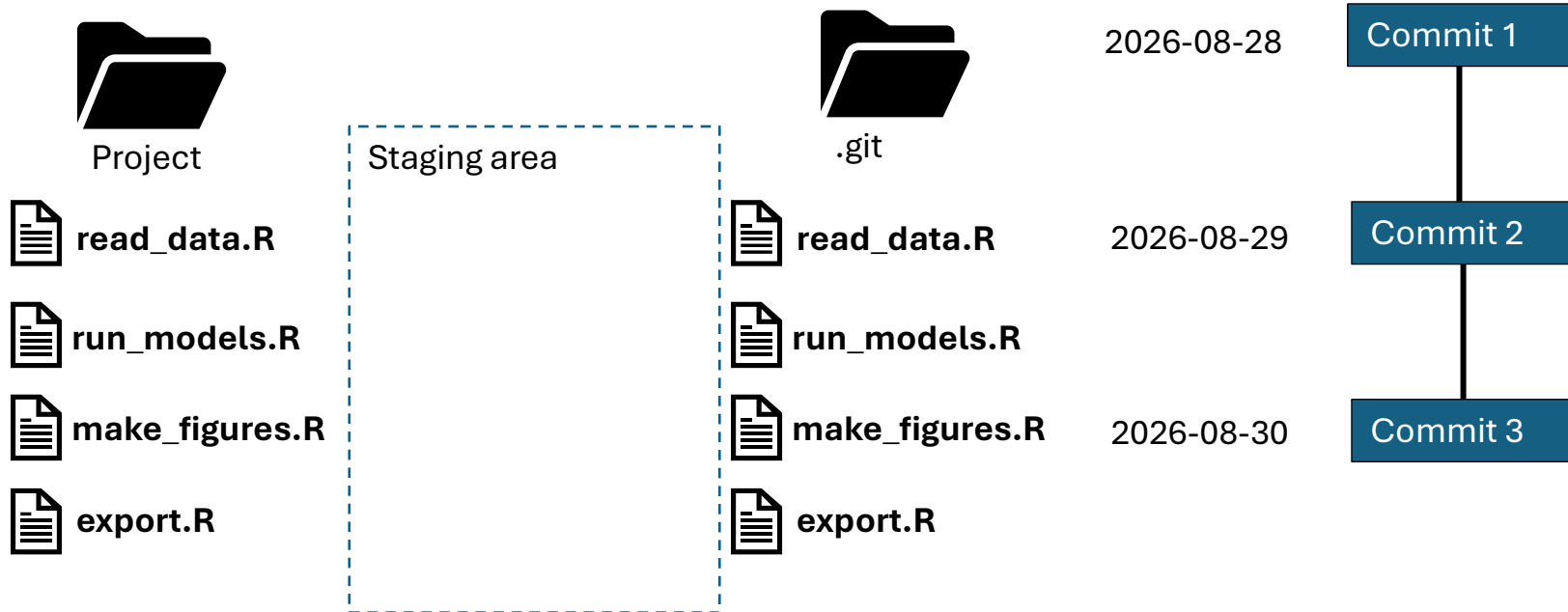
```
andreas ~ % git add make_figures.R
```

How does git work



```
andreas ~ % git commit -m "change color"
```

How does git work



Under the hood

- All versions of you project
- History always accessible



Project



read_data.R



run_models.R



make_figures.R



export.R



.git

```
.git
├── HEAD
├── config
├── description
├── hooks
│   ├── applypatch-msg.sample
│   ├── commit-msg.sample
│   ├── fsmonitor-watchman.sample
│   ├── post-update.sample
│   ├── pre-applypatch.sample
│   ├── pre-commit.sample
│   ├── pre-merge-commit.sample
│   ├── pre-push.sample
│   ├── pre-rebase.sample
│   ├── pre-receive.sample
│   ├── prepare-commit-msg.sample
│   ├── push-to-checkout.sample
│   ├── sendemail-validate.sample
│   └── update.sample
├── info
│   └── exclude
├── objects
│   ├── info
│   └── pack
│       ├── pack-cb15579e4b4f49688f4b737446f172c5ca4f5449.idx
│       ├── pack-cb15579e4b4f49688f4b737446f172c5ca4f5449.pack
│       └── pack-cb15579e4b4f49688f4b737446f172c5ca4f5449.rev
├── packed-refs
├── refs
│   ├── heads
│   └── tags
```

9 directories, 22 files

Writing a good commit message

What and why

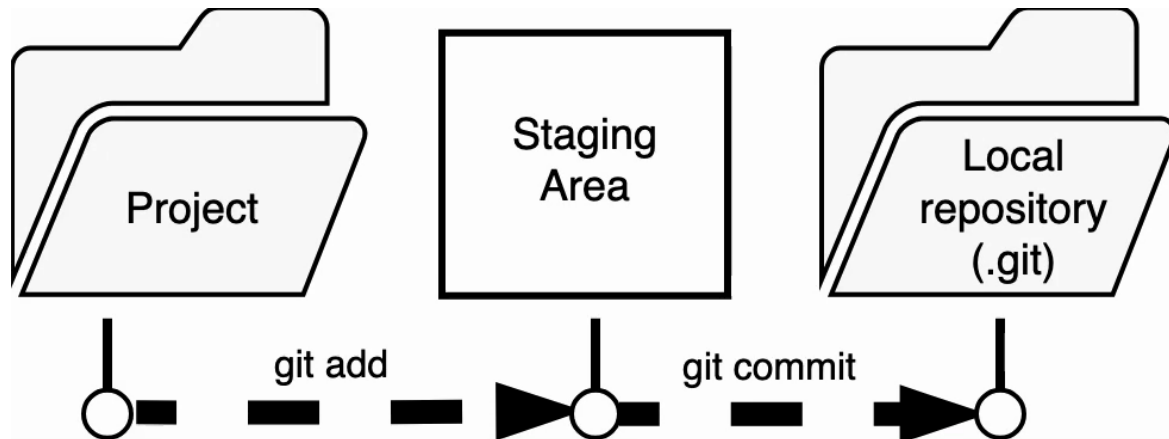
~~Updates~~

~~This change was done to describe
the essence of the code, in
particular lines 34-65~~

Fix na.rm bug in mean
calculations ✓



Summary



```
andreas ~ %  
  
git init      # start watching project  
  
git status    # check status of your project  
  
git add       # add files to track  
  
git commit    # store snapshot of project
```

References

Chacon, S. (with Straub, B.). (2014). *Pro Git* (2nd ed). Apress L. P.